Experiment 5: Fourier Series Analysis of Periodic Signals

OBJECTIVE 1:

To analyze Fourier series of continuous-time periodic signals and analyze their waveform Characteristics.

Experimental Procedure:

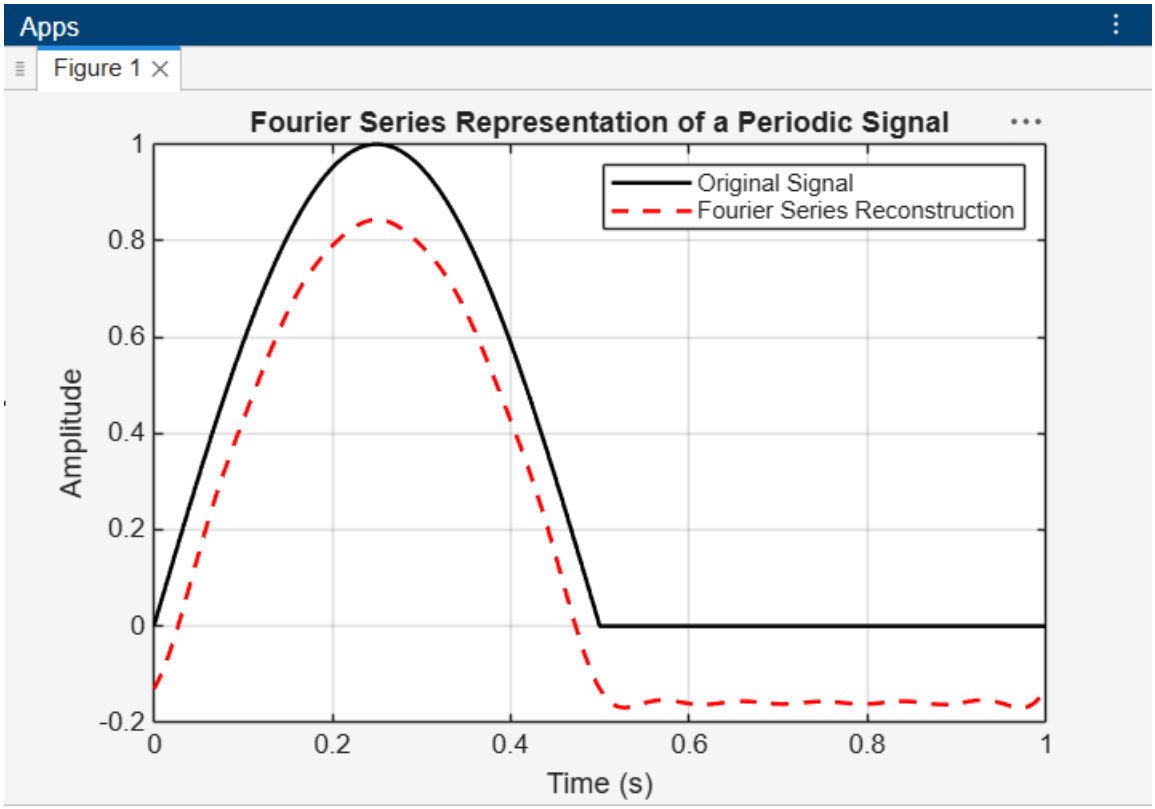
1. Define the time vector and fundamental frequency.
2. Generate the half-wave rectified sinusoidal periodic signal.
3. Compute the Fourier coefficients up to N = 10.
4. Construct the Fourier-series approximation.
5. Plot the original and reconstructed signals and compare them.

CODE:

```
clc;
clear;
close all;
% Fourier Series Representation
A = 1;
f0 = 1;
T = 1/f0;
t = linspace(0,T,2000);
% Half-wave rectified sinusoidal signal
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 10;
% Fourier coefficients
a0 = (1/T)*trapz(t,x);
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
```

```
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end
% Fourier Series reconstruction
xFS = a0/2*ones(size(t));
for k = 1:N
    xFS = xFS + an(k)*cos(2*pi*k*f0*t) ...
            + bn(k)*sin(2*pi*k*f0*t);
end
% Plotting
figure;
plot(t,x,'k','LineWidth',1.5);
hold on;
plot(t,xFS,'r--','LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Fourier Series Representation of a Periodic Signal');
legend('Original Signal','Fourier Series Reconstruction');
```

output:



Objective 4:

To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal

Experimental Procedure:

1. Generate the periodic signal over two periods.

2. Compute Fourier coefficients from one period.

3. Construct the reconstructed signal using 10 harmonics.

4. Calculate the error signal.

5. Plot the original, reconstructed and error signals.

Code:

```
clc;

clear;

close all;

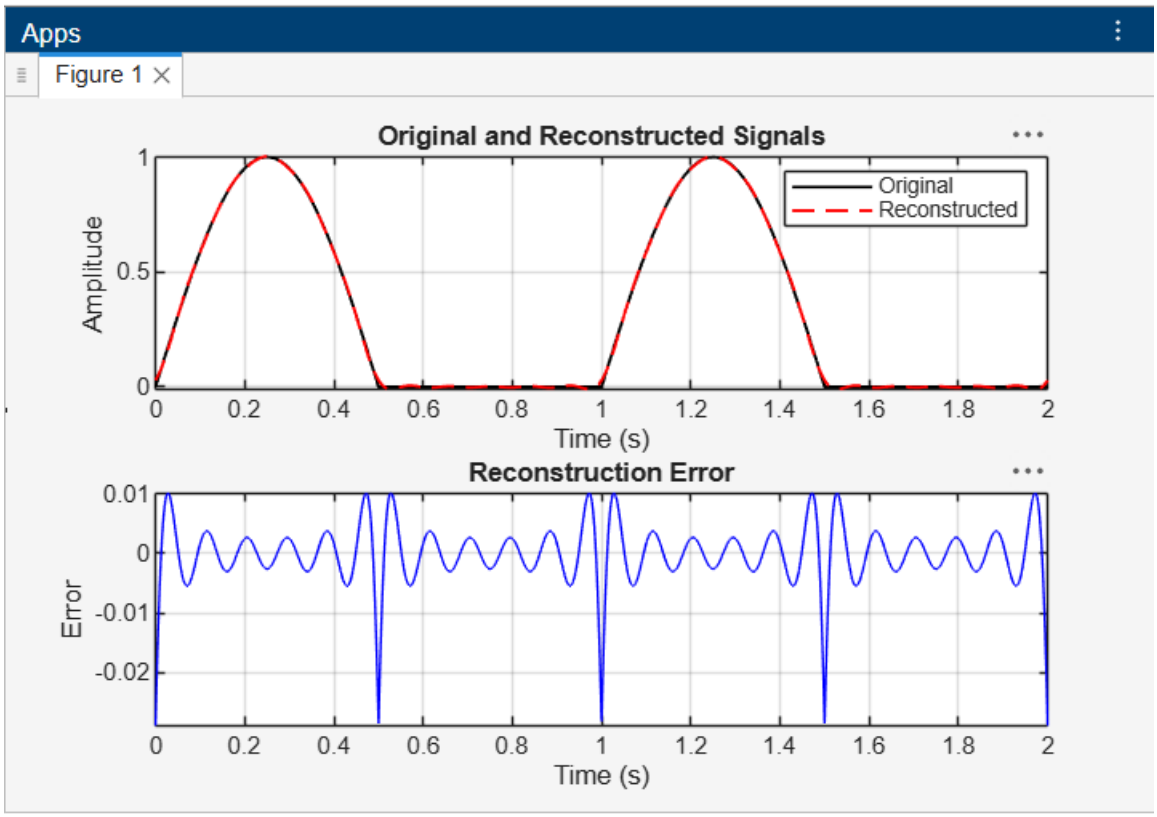
% Fourier Series Reconstruction and Error Analysis

f0 = 1;
```

```
T = 1/f0;
t = linspace(0,2*T,4000);
% Original signal
x = max(sin(2*pi*f0*t),0);
N = 10;
% Use one complete period
tt = t(1:2000);
xx = x(1:2000);
% Fourier coefficients
a0 = (2/T)*trapz(tt,xx)/2;
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N    an(k) = (2/T)*trapz(tt,xx.*cos(2*pi*k*f0*tt));
   bn(k) = (2/T)*trapz(tt,xx.*sin(2*pi*k*f0*tt));
end
% Fourier Series reconstruction
xr = a0*ones(size(t));
for k = 1:N
   xr = xr + an(k)*cos(2*pi*k*f0*t) ...
         + bn(k)*sin(2*pi*k*f0*t);
end
% Error calculation
err = x-xr;
% Plotting
figure;
subplot(2,1,1);
plot(t,x,'k',t,xr,'r--','LineWidth',1.1);
grid on;
```

```
xlabel('Time (s)');
ylabel('Amplitude');
title('Original and Reconstructed Signals');
legend('Original','Reconstructed');
subplot(2,1,2);
plot(t,err,'b');
grid on;
xlabel('Time (s)');
ylabel('Error');
title('Reconstruction Error');
```

output:



Result:

Thus, the Fourier Series analysis of continuous-time periodic signals was successfully performed. The Fourier coefficients were calculated, the original signal was reconstructed, and the reconstruction error was analyzed.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks (10)	Marks obtained
(i) To analyze the Fourier series of continuous-time periodic signals and analyze their waveform characteristics.	**Fourier Series Analysis**	Correctly determines the fundamental period and frequency and computes the Fourier coefficients for every given periodic signal, analytically or in MATLAB. Uses suitable commands with correct limits and obtains accurate coefficient values.	Computes the coefficients for most signals correctly, with minor errors in limits, algebra, code or parameters.	Computes the coefficients for only some signals, or has noticeable errors in integration, limits or code, and needs guidance.	Unable to compute the Fourier coefficients of the given signals.	**20**	2	

	Waveform Characteristics & Spectrum Analysis	Correctly identifies symmetry (even, odd, half-wave), DC value and the harmonics present. Plots the amplitude and phase spectra with titles, axis labels and grid. Explains how symmetry and the shape of the waveform determine the harmonic content and the rate of decay of the coefficients.	Identifies most waveform characteristics and plots the spectra correctly, with minor errors or limited explanation.	Identifies basic characteristics. Spectra have noticeable errors or the explanation is incomplete, and needs guidance.	Unable to identify the waveform characteristics or plot the spectra.	**20**	2	
(ii) To reconstruct periodic signals using the Fourier series and verify	**Signal Reconstruction using Fourier Series**	Correctly synthesizes each periodic signal as a partial sum of N harmonics in MATLAB, using a suitable time	Reconstructs the signals correctly with minor errors in code, time range, paramete	Reconstructs only some signals or uses a single value of N, with noticeable errors in code or	Unable to reconstruct the signals using the Fourier series.	**20**	2	

the reconstructed waveform with the original signal.		vector and loop or vectorized code. Reconstructs with several values of N and plots them clearly against the original.	rs or labelling, and uses fewer values of N.	plots, and needs assistance.				
	Verification of Reconstructed Waveform	Verifies the reconstruction against the original waveform using overlay plots and numerical error for each N. Shows that the error decreases as N increases, and identifies the Gibbs phenomenon at discontinuities with a correct explanation.	Verifies the reconstruction with overlay plots and some error measure, with minor errors in calculation or interpretation of convergence.	Performs basic visual comparison only, with noticeable errors or no numerical error. Convergence or Gibbs phenomenon is not explained, and needs guidance.	Unable to verify the reconstructed waveform against the original signal.	**20**	2	

Interpret the Fourier series results and maintain the record.	**Engineering Analysis, Record & Interpretation**	Provides technically accurate interpretation of the results, relating the number of harmonics, symmetry and discontinuities to reconstruction accuracy. Maintains a complete, well-organized record with code, plots, observations and conclusions.	Maintains a proper record with mostly accurate interpretation, with minor omissions in observations or conclusions.	Record and interpretation are basic or partly incomplete, with limited technical reasoning.	Incomplete record, limited or incorrect interpretation of the results.	**20**	2	
					Total	**100**	**10**	